

TITANIC

DATA ANALYSIS USING PYTHON

A Data Science Mini Project

TOOLS USED



PYTHON



PANDAS



MATPLOTLIB



DATASET

Titanic-Dataset.csv



PROJECT BY

Disha Bundela



DATE

27 May 2026



"Not even God can sink this ship."

– Titanic

TITANIC DATASET ANALYSIS USING PYTHON

By – Disha Bundela

TOOLS USED:

PYTHON

PANDAS

MATPLOTLIB

VS CODE

PRoJECT OVERVIEW

- **Objective**

- The goal of this project is to:
- Load and understand the Titanic dataset
- Clean missing values
- Analyze passenger information
- Visualize important insights using graphs

DATASET INFORMATION

- **Dataset Information**

- The Titanic dataset contains details about passengers who traveled on the Titanic ship. It is used for data analysis and machine learning to study survival patterns.

- **Key Information:**

- **Total Records:** 891 passengers
- **Total Columns:** 12 features
- **Data Types:** Integer, Float, and String values
- **Purpose:** Analyze factors affecting passenger survival

- **Important Columns:**

- **PassengerId** → Unique ID of each passenger
- **Survived** → Survival status (0 = No, 1 = Yes)
- **Pclass** → Passenger class (1st, 2nd, 3rd)
- **Name** → Passenger name
- **Sex** → Gender of passenger
- **Age** → Age of passenger
- **SibSp** → Number of siblings/spouses aboard
- **Parch** → Number of parents/children aboard
- **Ticket** → Ticket number
- **Fare** → Ticket fare amount
- **Cabin** → Cabin number
- **Embarked** → Port of embarkation

- **Observation:**

- Some columns such as **Age**, **Cabin**, and **Embarked** contained missing values.
- Data cleaning was performed to handle these missing values before visualization and analysis.

DATA CLEANING

- **Missing Values Found**

- Age → 177 missing values
- Cabin → 687 missing values
- Embarked → 2 missing values

- **Cleaning Performed**

- Filled Age missing values using mean
- Filled Embarked missing values using mode
- Removed/handled Cabin missing values

MISSING VALUES AFTER CLEANING

- **Result**
- After preprocessing:
- All missing values became 0
- Dataset became clean and ready for analysis
- **Benefits**
- Better analysis
- Accurate visualization
- Improved data quality

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0
dtype:	int64

```
Survived
0      549
1      342
Name: count, dtype: int64
```

```
Sex
male      577
female    314
Name: count, dtype: int64
PS C:\Users\Disha\Desktop>
```

[illegible]

SURVIVAL ANALYSIS

Status

Count

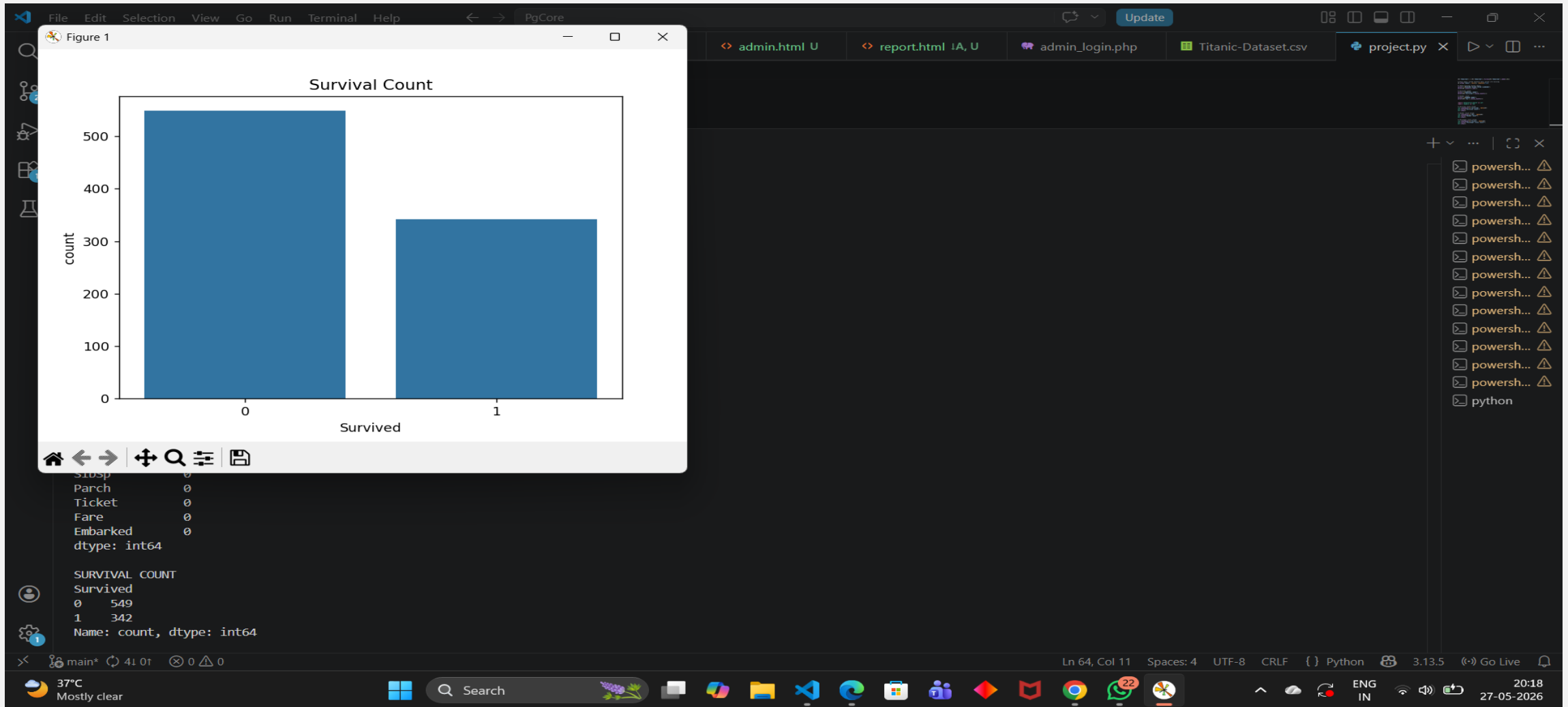
Did Not Survive (0)

549

Survived (1)

342

SURVIVAL COUNT GRAPH



GENDER ANALYSIS

Gender

Count

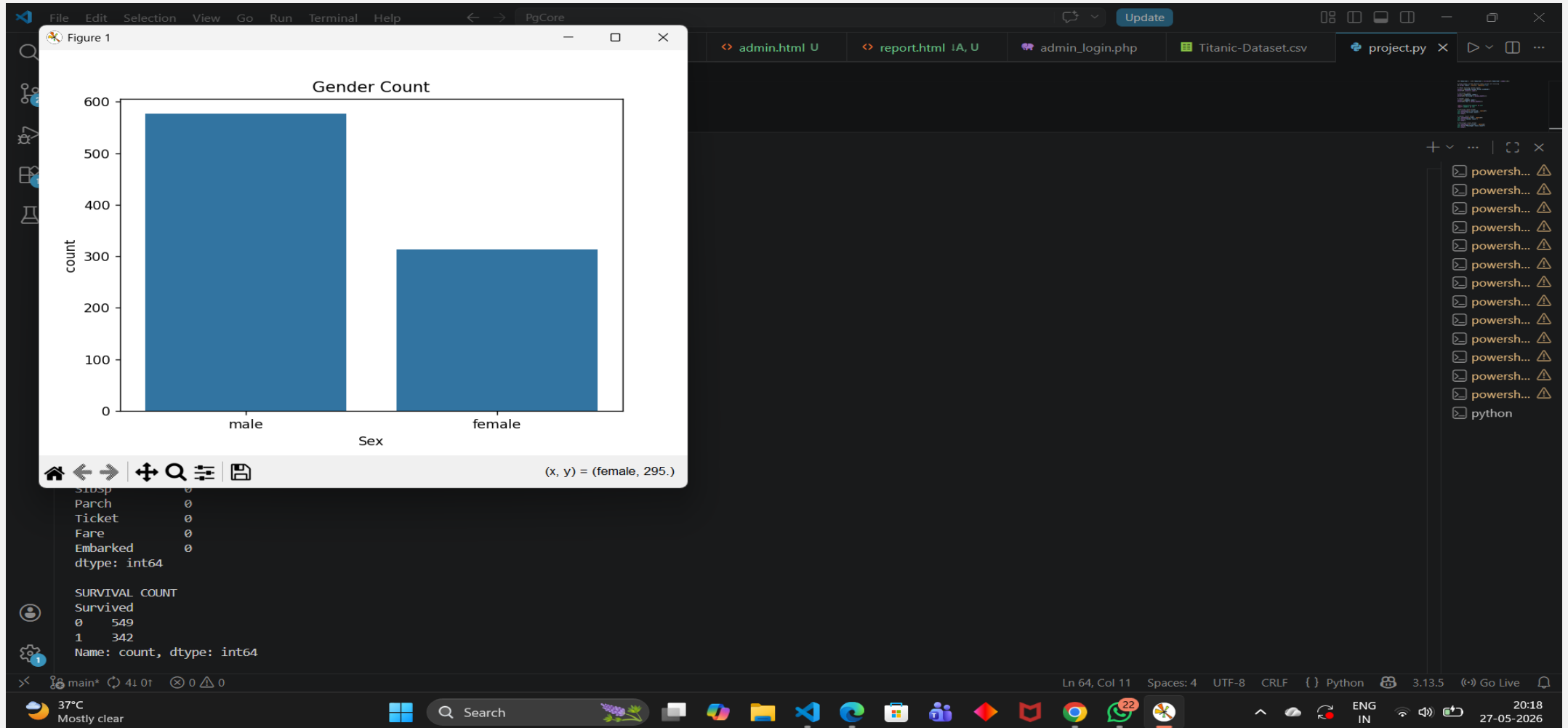
Male

577

Female

314

GENDER COUNT GRAPH



PASSENGER CLASS ANALYSIS

Class

Count

1st Class

216

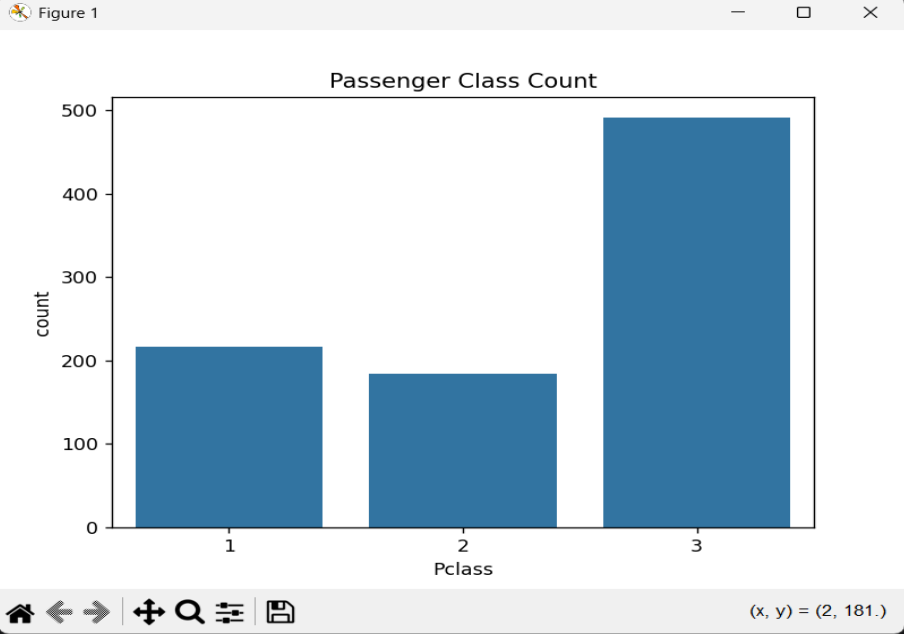
2nd Class

184

3rd Class

491

PASSENGER CLASS GRAPH



```
ticket      0
Fare        0
Embarked    0
dtype: int64

SURVIVAL COUNT
Survived
0      549
1      342
Name: count, dtype: int64
```

TECHNOLOGIES USED

- **Libraries Used**

- Pandas → Data handling
- Matplotlib → Data visualization

- **IDE Used**

- Visual Studio Code

CONCLUSION

- This project helped in understanding:
- Data loading
- Data cleaning
- Exploratory Data Analysis (EDA)
- Visualization techniques
- The Titanic dataset provided useful insights about passenger survival, gender distribution, and passenger classes.

THANK YOU!!

THANK YOU!

Thank you for your time and attention.
I hope you found this presentation
informative and insightful.



This project helped me understand
data analysis, data cleaning,
and visualization using Python
libraries like Pandas and Matplotlib.

PRESENTED BY

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27 May 2026



India



Have a great day!